

Sequential Read/Write: This test is conducted with a single large file, so the data is 'in sequence'. The hard disk already knows where to look for the data, and hence this results in faster data transfer. If you transfer a single large file of, say, 1 GB, instead of multiple smaller files totalling up to 1 GB, the single file will be copied faster.

Random Read/Write: This is conducted with multiple smaller files instead of a single large file, and the drive has to search for the next file every time it is done copying another. This test, therefore, results in slower times clocked.

Average Access Time: The interval between the time the request for data is made by the system and the time the data is available off the drive.

CPU Utilisation: Since a copying or backup job would probably be going on in the background and not as a dedicated process, the drive that utilises less CPU time is a better drive. It will give you better system performance. Lower CPU utilisation also means that the drive can be attached to older, lower-end processors without the system hanging due to lack of availability of computing resources.

The tests we described so far were synthetic tests—they measure drive performance reasonably well. But they are called synthetic for a reason; they might or might not reflect real-world usage. So we also did a good old copy-pasting test, and copied 1 GB of assorted data to and from the drive, as well as single 1 GB file to test the sequential and random read/write parameters.

Now that you have a basic understanding of the tests, let's get straight to the tests themselves.

PORTABLE HARD DISK DRIVES

Looks, Size and Build

We received drives in various shapes and sizes, from circular to rectangular to square. The only shapes missing were triangles and ellipses!

The Seagate pocket drives were just that—pocketable. They were scarcely bigger in circumference than a cookie: they were 1-inch drives and were quite thin as well.

But all they could hold was a mere 2.5 and 5 GB.

In this segment, the largest capacity was that of the Seagate 100 GB. This was a cool-looking square drive that had a metallic mesh. It was slightly bulky and somewhat heavy—not exactly a 'pocket' drive.

A really light and pocketable drive was the Freecom FHD2 Pro 60 GB. This drive is wafer-thin and extremely light. You wouldn't feel it if you were carrying it in your pocket, though the pocket would need to be large—the drive is rectangular and rather long.

The Western Digital (WD) 40 GB Passport was low on storage space, and quite bulky. But it featured a rubberised underbody that ensured that vibrations and noise—which are, in any case, next to nothing—were minimised, and that the disk stayed put on any surface. The ports are protected by a rubber flap that resembles the port concealers on everyday digicams.

If you are looking for something that's stylish with lots of space, and is not just portable but actually pocketable, look no further than the Transcend Storejet drives. They come in 20 and 40 GB models, and are smaller than a deck of cards. They weigh just 100 grams!

In our portable HDD shootout, the only drive that came with the one-touch backup feature was the Freecom FHD Pro 2 60 GB.

Funnily enough, in terms of visual indicators, all the drives preferred a cool blue for the LED, or even on the buttons (as in the case of the Maxtor and the WD). That could be an oblique reference to the blues of backup!

On To The Tests

Synthetic Benchmarking: We started with the SiSoft Sandra file system benchmark. The drive that absolutely blazed through the synthetic benchmarks was the Seagate 100 GB. It returned a drive index of 28. The only hitch, performance-wise, was that the CPU utilisation was marginally higher, at 3.75 per cent. Remember that the CPU we were using is extremely fast, so CPU utilisation on lower-end CPUs would be a lot more than this.

This drive could put up such a fantastic show because of its 8



Freecom FHD2 - Pro 60 GB

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